



INDIAN INSTITUTE OF SCIENCE, BANGALORE

COURSE PROJECT: CP230 Motion Planning for Autonomous Systems

Efficient Path Planning for Quadrotor/Mobile robots using RRT with Safety certificates

Bonus Problem: Path Planning for Multi-agent system deployed to carry a fragile object to avoid Static Obstacles and reach Static Goal

Paper: Efficient collision checking in sampling-based motion planning via safety certificates
Joshua Bialkowski, Michael Otte, Sertac Karaman and Emilio Frazzoli

Group 10: 1) Lokesh Bansal 2) Vishwas BC 3) Suprakash Saren

Teaching Assistant: Mr. Amit Kumar



Efficient collision checking in sampling-based motion planning via safety certificates

- Computational bottleneck (collision checking) in sampling-based motion planning algorithms.
- Safety check has helped to reduce the computational complexity for collision checking.
- Expected time spent on collision checking per iteration approaches zero (as the number of iterations approaches infinity)

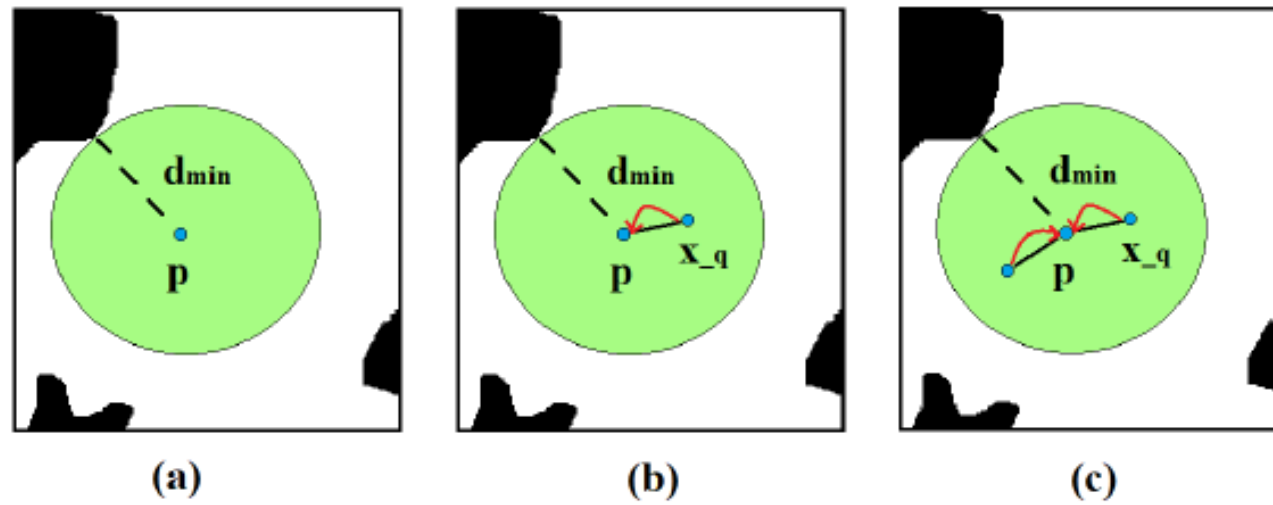


Fig. 1: (a) “Safety certificates” (green) defined by d_{min} (Radius denoted as dashed black lines) for point p . (b) Next sampled node, x_q , is within a certificate need not have collision checking. (c) Pointers (solid red arrows) are maintained to certifying nodes. used

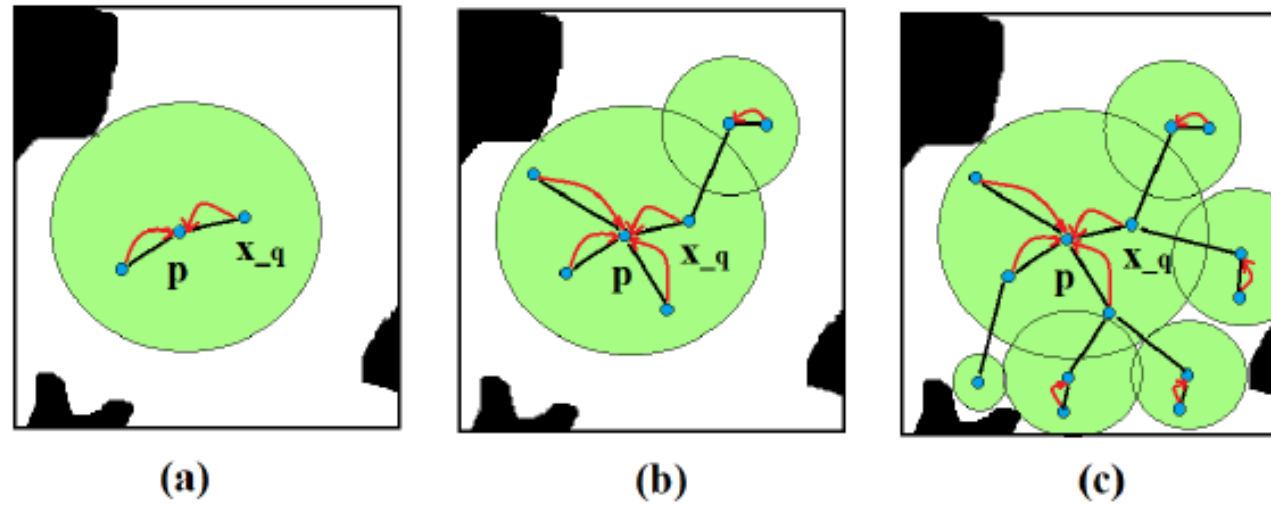
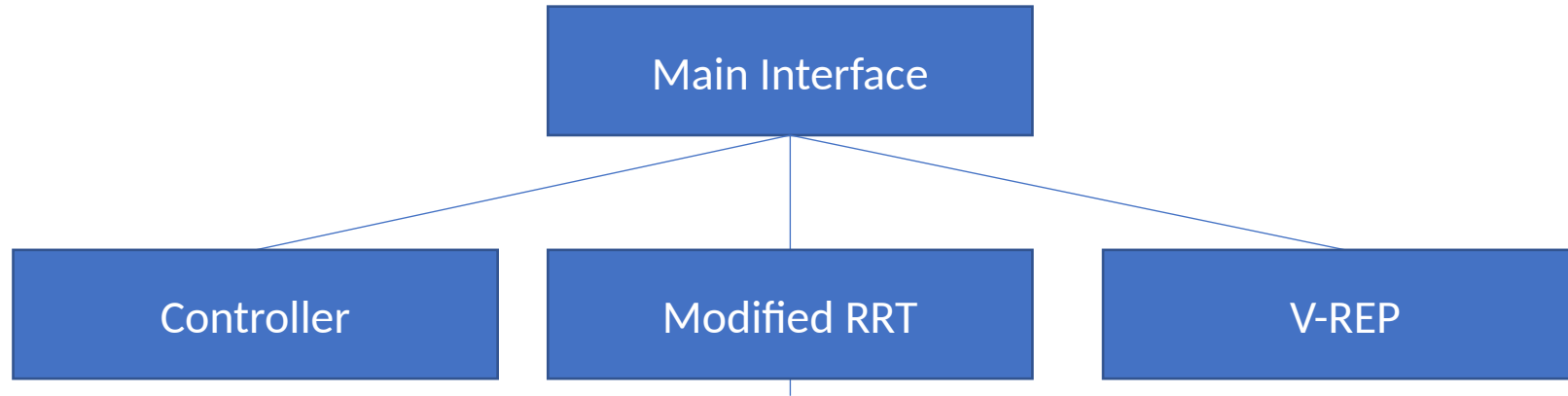


Fig. 2: Certificates (green discs) asymptotically cover the space as graph size increases toward infinity. The ratio of collision checks versus graph size (nodes) approaches zero in the limit as graph size approaches infinity.

Implementation



- Modified RRT
 - Give a trajectory from start to goal
 - The waypoints of the trajectory are stored as csv file and would be called by main function
- Controller
 - PD controller to track the trajectory given by the Modified RRT
- V-REP
 - An environment with obstacles is created
 - Connect V-REP to MATLAB

Algorithm 1: CFreePoint(x_q)

```
1  $x_{\text{free}} \leftarrow \text{Nearest}(S_{\text{free}}, x_q)$ ;  
2  $x_{\text{obs}} \leftarrow \text{Nearest}(S_{\text{obs}}, x_q)$ ;  
3 if  $\|x_q - x_{\text{free}}\| \leq \text{Dist}(x_{\text{free}})$  then  
4    $\text{CertifierOf}(x_q) \leftarrow x_{\text{free}}$ ;  
5   return True  
6 else if  $\|x_q - x_{\text{obs}}\| \leq \text{Dist}(x_{\text{obs}})$  then  
7   return False  
8  $d_{\text{obs}} \leftarrow \text{SetDistance}(X_{\text{obs}}, x_q)$ ;  
9 if  $d_{\text{obs}} > 0$  then  
10   $S_{\text{free}} \leftarrow S_{\text{free}} \cup \{x_q\}$ ;  
11   $\text{Dist}(x_q) \leftarrow d_{\text{obs}}$ ;  
12   $\text{CertifierOf}(x_q) \leftarrow x_q$ ;  
13  return True  
14 else  
15   $S_{\text{obs}} \leftarrow S_{\text{obs}} \cup \{x_q\}$ ;  
16   $\text{Dist}(x_q) \leftarrow \text{SetDistance}(X_{\text{free}}, x_q)$  ;  
17  return False
```

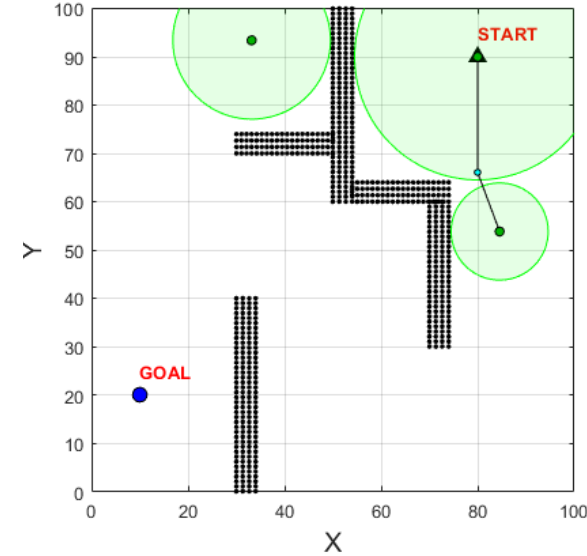
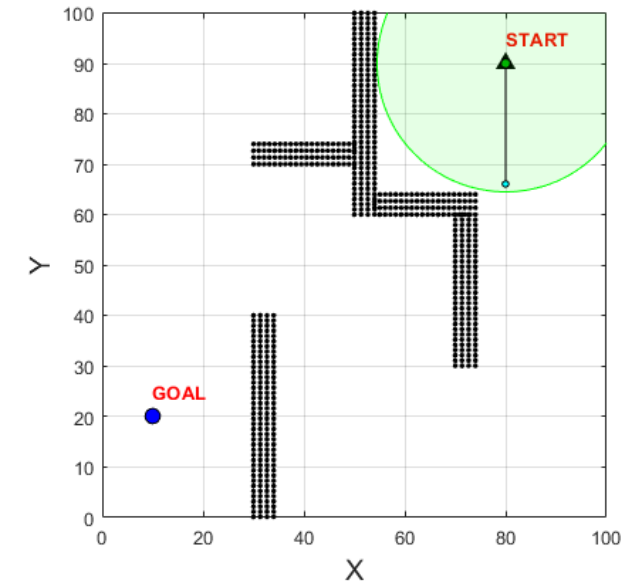
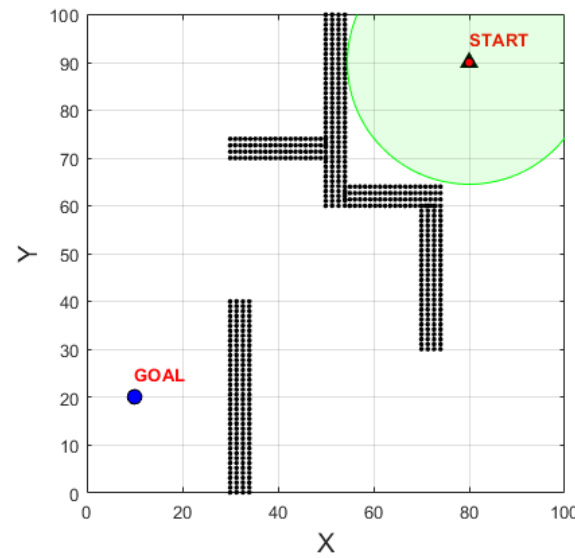


Fig.3 Step-by-step implementation of algorithm 1

Algorithm 2: BatchCFreePath(S_{near}, x_q)

```
1  $H \leftarrow \emptyset$ ;  
2  $c_q \leftarrow \text{CertifierOf}(x_q)$ ;  
3 foreach  $x \in S_{\text{near}}$  do  
4   if  $\|x - c_q\| \leq \text{Dist}(c_q)$  then  
5      $H \leftarrow H \cup \{(x, x_q)\}$ ;  
6   else  
7      $c \leftarrow \text{CertifierOf}(x)$ ;  
8     if  $\|\text{FarSafe}(x_q, x) - c\| \leq \text{Dist}(c)$  then  
9        $H \leftarrow H \cup \{(x, x_q)\}$ ;  
10    else if  $\text{CFreePath}(x, x_q)$  then  
11       $H \leftarrow H \cup \{(x, x_q)\}$ ;  
12 return  $H$ 
```

Algorithm 3: Modified RRT

```
1  $V \leftarrow \{x_{\text{init}}\}$ ;  $E \leftarrow \emptyset$ ;  $S_{\text{free}} \leftarrow \emptyset$ ;  $S_{\text{obs}} \leftarrow \emptyset$ ;  
2 for  $i = 1, \dots, n - 1$  do  
3    $x_{\text{rand}} \leftarrow \text{Sample}_i(\omega)$ ;  
4   if  $\text{CFreePoint}(x_{\text{rand}})$  then  
5      $x_{\text{nearest}} \leftarrow \text{Nearest}(V, x_{\text{rand}})$ ;  
6      $H \leftarrow \text{BatchCFreePath}(\{x_{\text{nearest}}\}, x_{\text{rand}})$ ;  
7     if  $H \neq \emptyset$  then  
8        $V \leftarrow V \cup x_{\text{rand}}$ ;  
9        $E \leftarrow E \cup H$ ;  
10 return  $G = (V, E)$ 
```

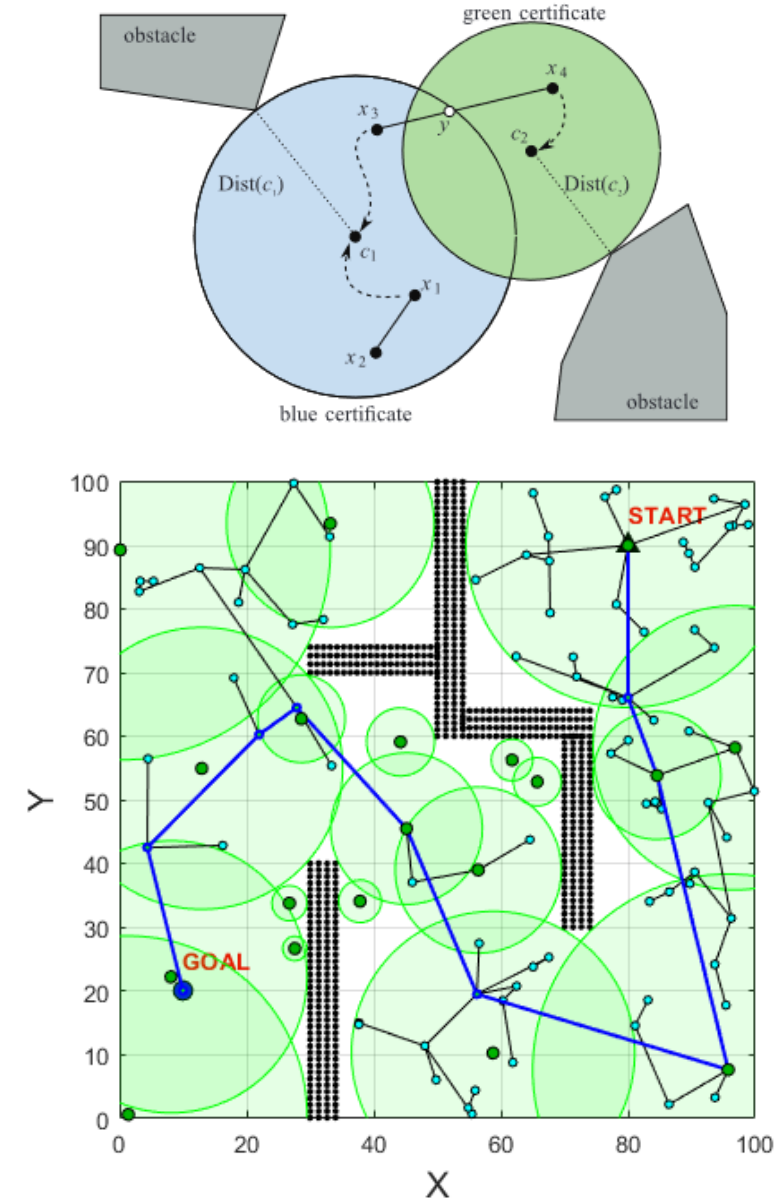


Fig.4 Step-by-step implementation of algorithm 2 and 3

Path Planning for Quadrotor: RRT with Safety Certification

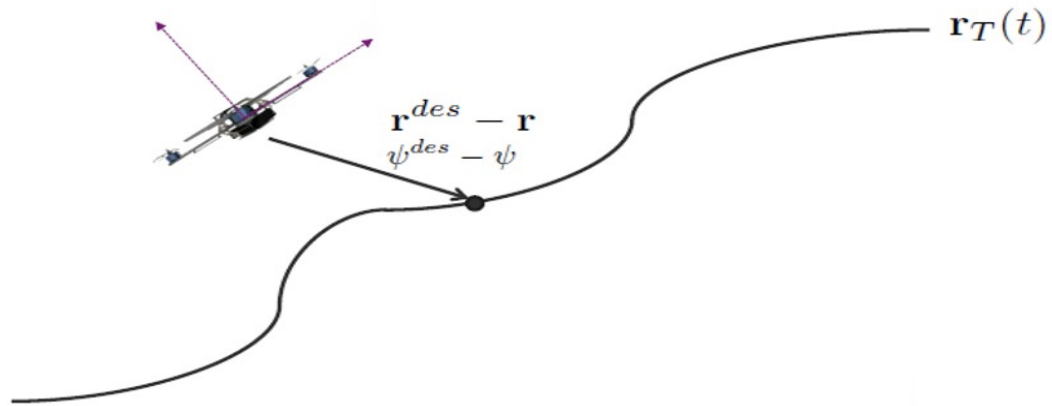


Figure 5: PD Controller for trajectory tracking with 6DOF Quadrotor Model

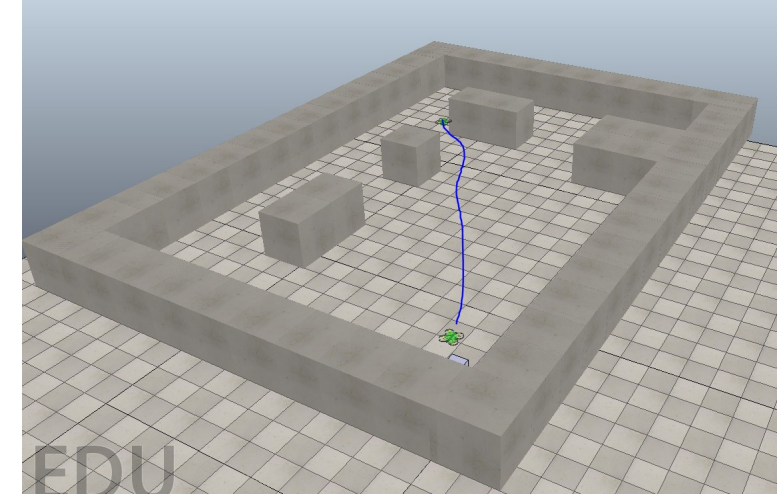


Figure 6: Avoid Static obstacles and reach a static target

Objective

- Implement **RRT with Safety Certification** for **collision checking** to generate trajectory by **avoiding static obstacles** and **reach static target**
- Implement **6 DOF Quadrotor with PD Controller** for trajectory tracking for Quadrotor
- Simulation in **MATLAB** and **CoppeliaSim (V-REP)**

Modified RRT: Efficient Collision checking in node sampling

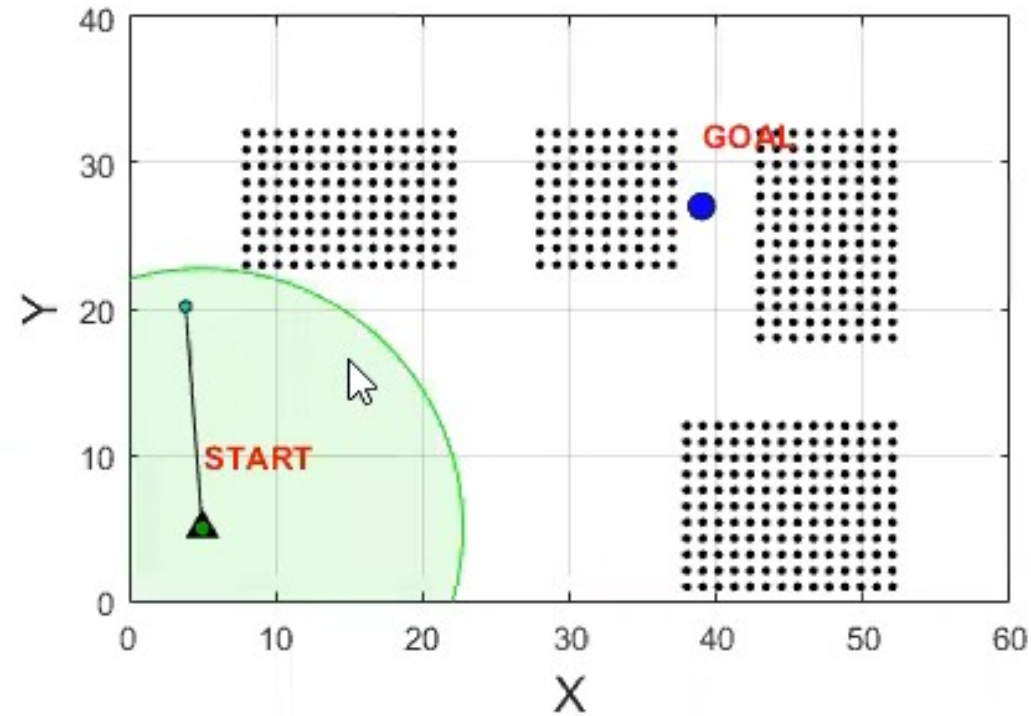


Figure 7: Efficient node sampling

- Green circles are the safety certificates
- If any newly sample node lies inside the green circle, it is implicitly declared to be collision free
- Explicit collision check need to perform if newly sample node doesn't lie inside the green circles

Execution time comparison of RRT(w/o SC) and Modified RRT with SC

SC: Safety Certification

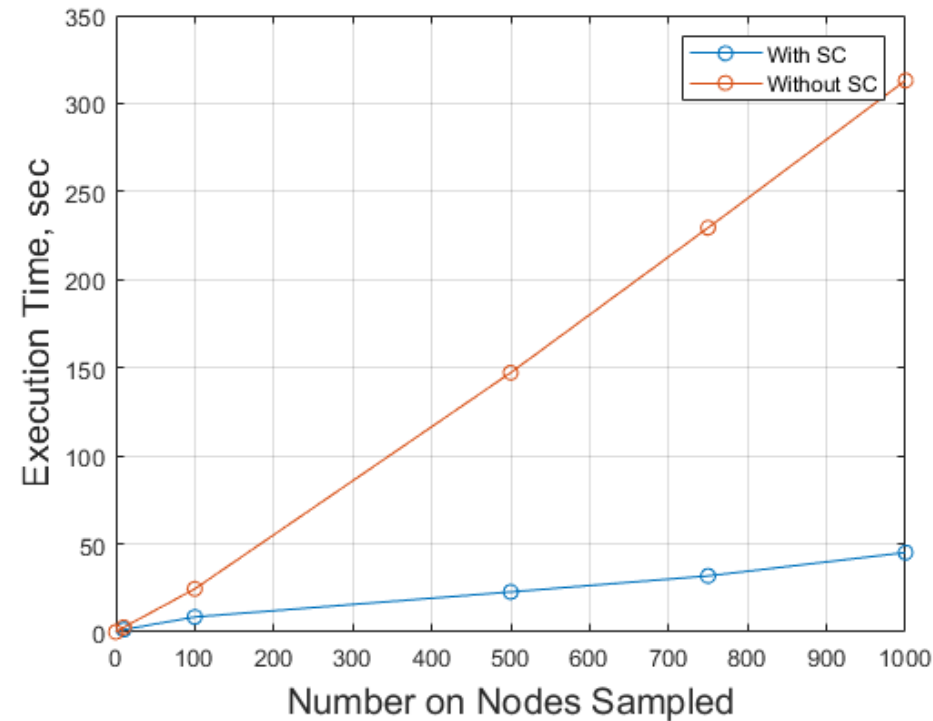


Figure 8: Efficient node sampling

- In absence of goal node:
 - 100, 500, 750 and 1000 nodes are sampled with SC and without SC
- Observed: Execution time of RRT algorithm reduced significantly

Path planning for Quadcopter with Modified RRT



Fig. 9: VREP Simulation with Quadcopter

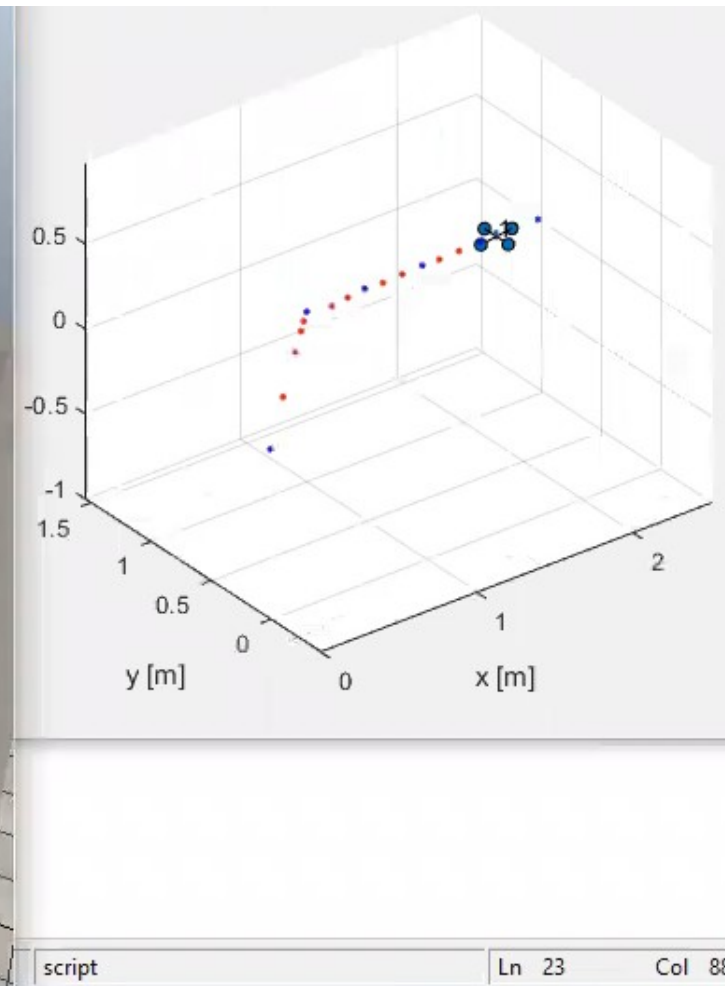


Fig. 10: Matlab Simulation with 6 DOF Quadcopter

Bonus Problem

Objective:

- The modified RRT algorithm is valid for a point mass system without considering the robot's dimension.
- This algorithm need to further modified to plan a collision free path for a multiagent system deployed to carry a fragile object of fixed dimension.
- The objective is to get a obstacle free path for the whole system (multiple robots + fragile object).

Environment Considered for the Problem:

- A grid map is assumed as a warehouse with static obstacles and goal location.
- In the grid two ground robots are deployed to carry a fragile object of largest dimension 'L'
- Goal and start locations are static and assumed to be collision free

Bonus Problem - Solution

- Two robots are chosen moving in a fixed formation to carry a fixed dimension object.
- The two robots can be placed, underneath the object with largest dimension L .
- We consider a circle of diameter ' L ' around the centroid of the formation.
- Now, we can modify the above algorithm as follows:
 - For drawing circles for safety certificates safe region is modified
 - Instead of taking the radius of safety certificate greater than 0, radius must be greater than L
 - If sampled node is not at least L unit away from an obstacle we consider this node to be invalid

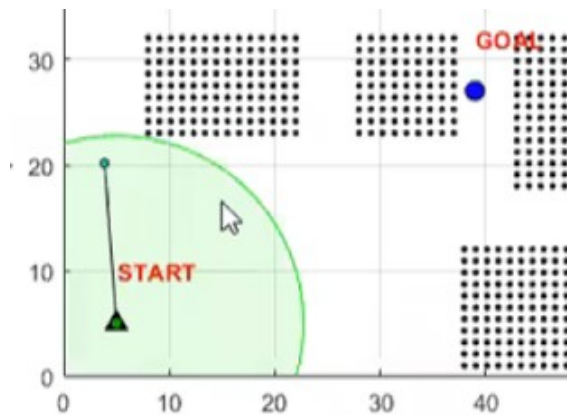


Fig. 11: Safety certificates are 0 distance apart from obstacles

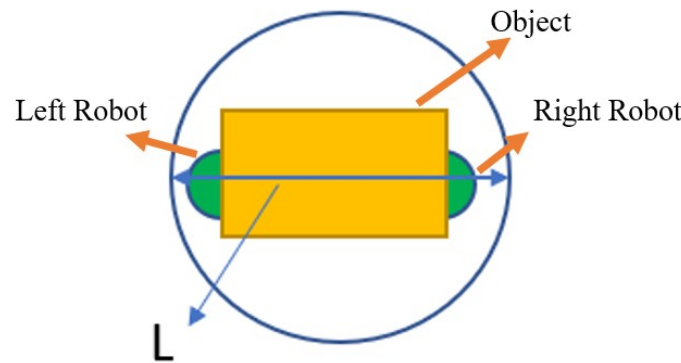


Fig. 12: Two robots carrying fragile object



Fig. 13: Safety certificates are L distance apart from obstacles

Bonus Problem - Solution

Algorithm 1: *ModifiedCFreePoint*(x_q)

```

1  $x_{free} \leftarrow \text{Nearest}(S_{free}, x_q);$ 
2  $x_{obs} \leftarrow \text{Nearest}(S_{obs}, x_q);$ 
3 if  $\|x_q - x_{free}\| \leq \text{Dist}(x_{free})$  then
4    $\text{CertifierOf}(x_q) \leftarrow x_{free};$ 
5   return True
6 else
7   if  $\|x_q - x_{obs}\| \leq \text{Dist}(x_{obs})$  then
8     return False
9   end
10 end
11  $d_{obs} \leftarrow \text{SetDistance}(X_{obs}, x_q);$ 
12 if  $d_{obs} > L$  then
13    $d_{obs} \leftarrow d_{obs} - L;$ 
14    $S_{free} \leftarrow S_{free} \cup x_q;$ 
15    $\text{Dist}(x_q) \leftarrow d_{obs};$ 
16    $\text{CertifierOf}(x_q) \leftarrow x_q;$ 
17   return True
18 else
19   if  $d_{obs} < \text{marginalvalue}$  then
20      $S_{obs} \leftarrow S_{obs} \cup x_q;$ 
21     return False
22   end
23 end
24 else  $\text{invalid} \leftarrow \text{invalid} \cup x_q;$ 

```

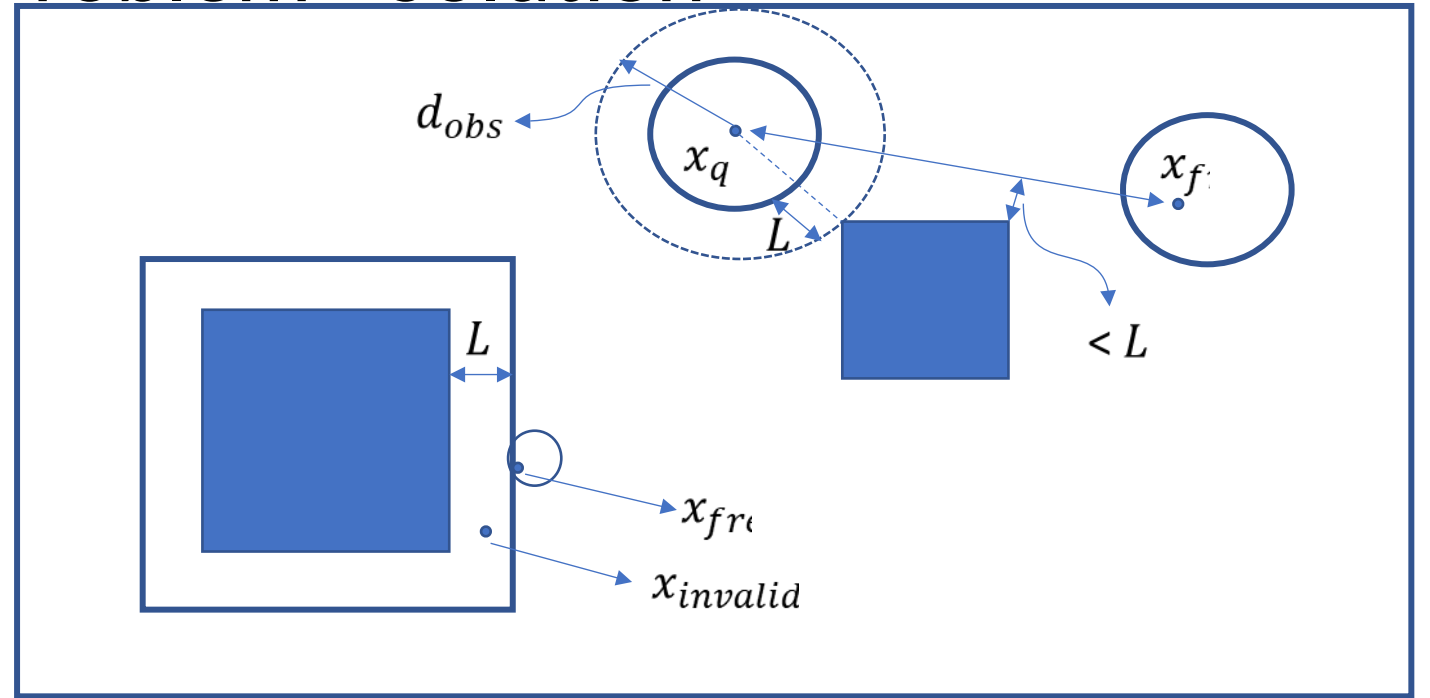


Fig. 14: Safety Certificates formation for multi-agent systems path planning

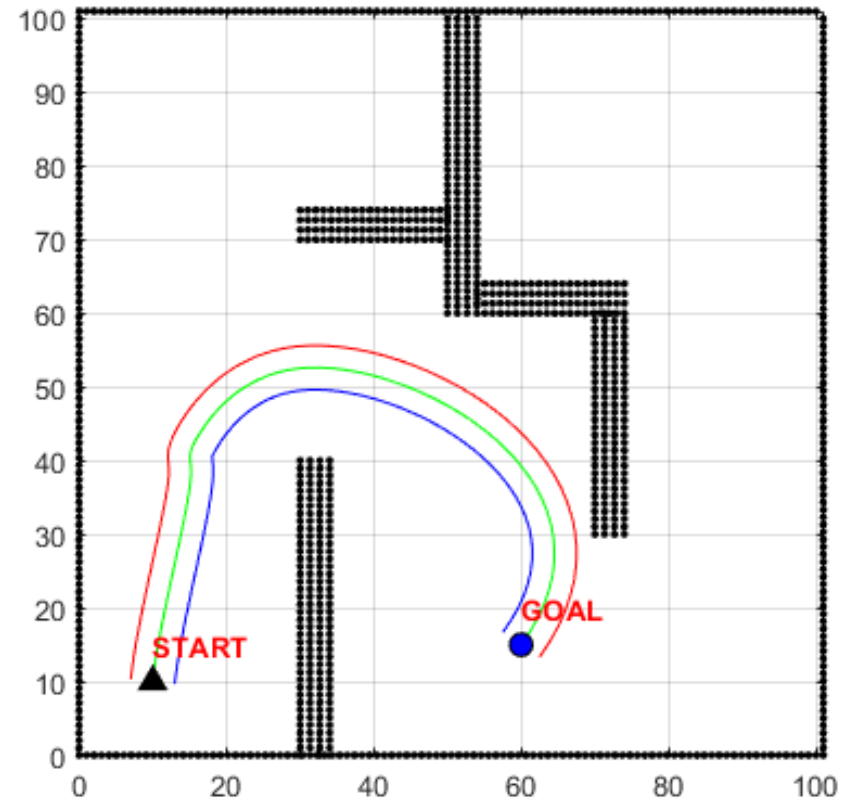
Algorithm 2: *ModifiedCFreePath*(x, x_q)

```

1  $\text{MinDistance} \leftarrow \text{argmin} \|(x, y) - x_{obs}\|;$ 
2 if  $\text{MinDistance} > L$  then
3    $H \leftarrow H \cup (x, x_q);$ 
4   return True
5 else
6   return False
7 end

```

Confidence Interval



1. The two robots can take any path in between red and blue paths
2. Green path is the path of the centroid of the multi-robot assembly, this path is generated by the modified RRT

Path Planning for Multi-robot assembly deployed to carry a fragile object to avoid Static Obstacles and reach Static Goal using RRT with Safety Certification

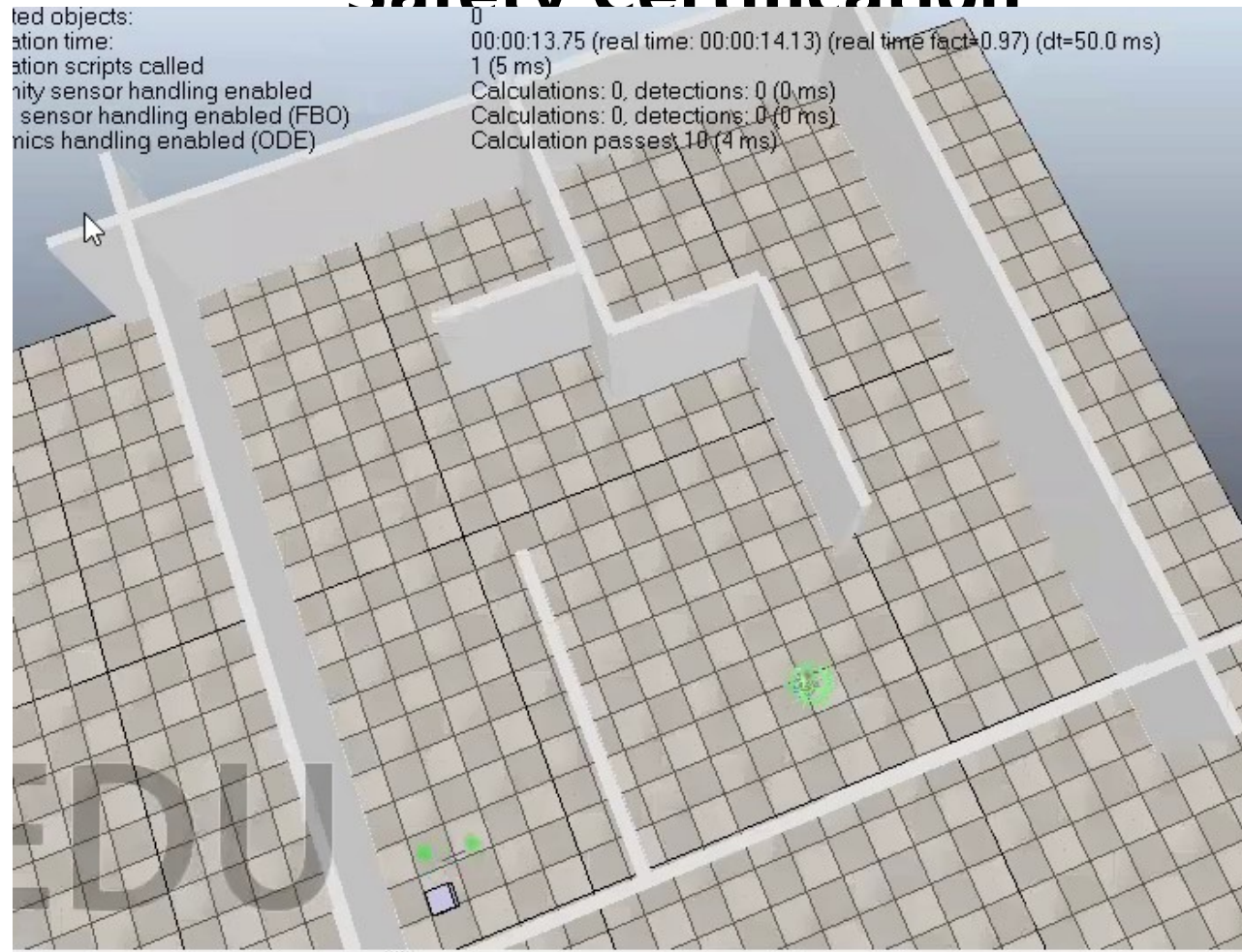


Fig. 16: VREP Simulation: two robots supposed to carry a fragile object

1. Modified RRT is used to give safe path from start to goal in static environment
2. Simulation CoppeliaSim (V-REP)

Thank You